INCH-POUND

MIL-PRF-1/334D
w/AMENDMENT 1
22 March 2017
SUPERSEDING
MIL-PRF-1/334D
6 March 2015

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING
TYPE 2C53

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein
shall consist of this document and the latest issue of MIL-PRF-1

DESCRIPTION: High-Mu triode.

Outline - - - See figure 1.

Base - - - B8-26.

Cap - - - C1-1.

Envelope - - - T9.

Cathode - - - Coated unipotential.

Base connections:

| Pin no. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | Cap |
|---|---|---|---|---|---|---|---|---|---|
| Element | sd and base sleeve | h | nc | nc | g | nc | h | k | a |

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | Ef | Eb | Ec | Pp | Ehk | Rk | Alt |
|---|---|---|---|---|---|---|---|
| Unit: | V | V dc | V dc | W | v | Ohms | ft |
| Maximum: | 6.9 | 8,000 | -200 | 12 | 300 | - - - | note 1 |
| Minimum: | 5.7 | - - - | - - - | - - - | - - - | - - - | note 1 |
| Test conditions: | 6.3 | 8,000 | 0 | - - - | 300 | 9,600 | - - - |

GENERAL:

Qualification – Not Required.

This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

AMSC N/A

FSC 5960

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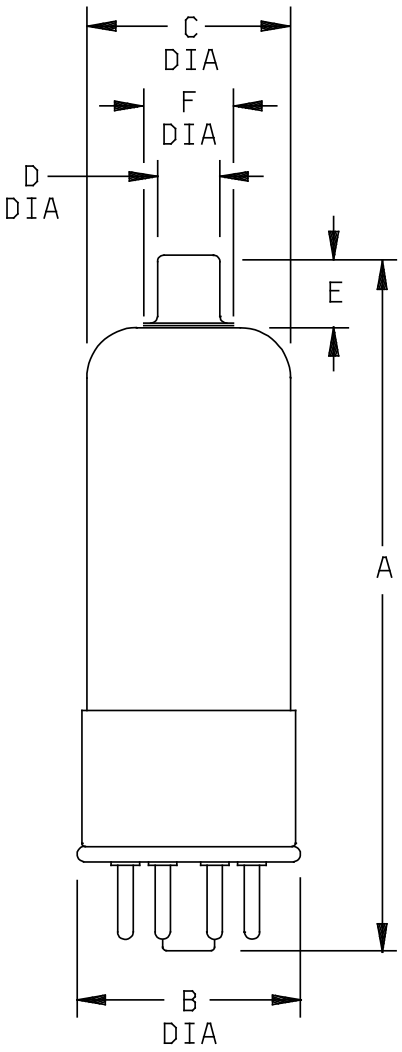
TABLE I. Testing and inspection.

| Inspection | Method MIL-STD-1311 | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|
| Conformance inspection, part 1 | | | | | | |
| Total grid current | 1266 | Eb = 6,000 V, see note 2 | Ic | 0 | -0.5 | μA dc |
| Emission | 1231 | Eb = Ec = 30 V dc, see note 2 | Is | 32 | - - - | mA dc |
| Electrode current (cathode) | 1256 | T = 60; Eb = 6,000 V, see note 3 | Ik | 0.9 | 1.6 | mA dc |
| Short and discontinuity detection | 1201 | | - - - | - - - | - - - | - - - |
| Conformance inspection, part 2 | | | | | | |
| Heater current | 1301 | | If | 275 | 325 | mA |
| Heater-cathode leakage | 1336 | | Ihk | - - - | 30 | μA dc |
| Transconductance | 1306 | Eb = 1,500 V dc; Ec = -1 V dc; Rk = 0 | Sm | 675 | 1,125 | μmhos |
| Amplification factor | 1316 | Eb = 1,500 V dc; Ec = -1 V dc; Rk = 0 | Mu | 380 | 520 | - - - |
| Direct-interelectrode capacitance | 1331 | No shield | Cgp<br>Cin<br>Cout | 0.60<br>4.0<br>1.2 | 0.90<br>6.0<br>2.6 | pF<br>pF<br>pF |
| Insulation of electrodes | 1211 | | - - - | - - - | - - - | - - - |
| Low-frequency vibration | 1031 | Eb = 300 V dc; Ec = 0; Rp = 10,000 Ω | Ep | - - - | 200 | mV ac |
| Base pin solder depth | 1111 | | - - - | - - - | - - - | - - - |
| Secureness of base, cap, or insert | 1101 | | - - - | - - - | - - - | - - - |
| Permanence of marking | 1105 | | - - - | - - - | - - - | - - - |
| Conformance inspection, part 3 | | | | | | |
| Life-test provisions | - - - | Group C; Ehk = 300 V | - - - | - - - | - - - | - - - |
| Life-test end point (500 hours) | - - - | Electrode current (cathode) | Ik | 0.75 | - - - | mA dc |

NOTES:
1. See “Reduced pressure (altitude) rating,” and altitude maximum peak voltage.
2. This test to be performed at the conclusion of the holding period.
3. Initial arcing may be tolerated, but tubes shall be free of arcing at the conclusion of the test.



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| Dimensions in inches with metric equivalents (mm) in parentheses. | | |
|---|---|---|
| Ltr | Minimum | Maximum |
| Conformance inspection, part 2 | | |
| A | 3.500 (88.90) | 3.875 (98.43) |
| B | | 1.312 (33.32) |
| Conformance inspection, part 3 (periodic-check) | | |
| C | 1.062 (26.97) | 1.188 (30.18) |
| D | 0.355 (9.02) | 0.365 (9.27) |
| E | 0.406 (10.31) Nominal | |
| F | 0.420 (10.67) Nominal | |

FIGURE 1. Outline drawing of electron tube type 2C53.



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Referenced documents. In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311.

The margins of this specification are marked with vertical line to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2017-005)

Review activities:
Army - MI
Navy - AS, CG, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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